Adding iNO in-line to HFPV

Set up iNO with standard Mallinckrodt connections—NO/NO_2 tubing and patient gas sample line with filter from injector module to monitor and injector module to DSIR monitor

1. Connect AeroNOx NO connector (orange tubing) to monitoring sample port on Phasitron
2. Connect AeroNOX NO connector (orange tubing) to iNO injection port of NO swivel tee
3. Connect iNO sample line to DSIR monitor and to AeroNOx NO connector (orange tubing)
4. Connect O_2 tubing to AeroNOx No connector (orange tubing) on the iNO injection port on Phasitron and the other end of the O2 tubing to the O2 flow meter/blender on isolette (flow will drive NO to bleed in-line to the patient
5. Connect another O_2 tubing from injector module (opposite end, after arrow pointing to the patient) to AeroNOx NO connector (orange tubing) on monitoring sample port on Phasitron

NOTE: **iNO delivery is based off a flow system: 2 – 5LPM (flow is directly proportional to amount of iNO being administered)**
NOTE: **A minimum flow of 2L needed for DSIR monitor to accurately measure iNO being injected into the circuit (on "Settings Screen" of DSIR monitor, a yellow and green reference bar graph measures the flow)**

1. To deliver 20ppm, set flow on flow meter on transport isolette to at least 2LPM and titrate to achieve desired dose to be administered
   a. Verify NO on DSIR monitor matches dosage amount ordered by provider
   b. To deliver 20 ppm of NO, first max out dosage to 80 ppm with a flow of 2 LPM
   c. If 20 ppm is still not being delivered per DSIR monitor, slowly increase flow, but do not exceed 5 LPM
   d. If NO being delivered is more than 20 ppm with a set dose of 80 ppm and a flow 2 LPM, slowly decrease set dose until 20 ppm is achieved
2. Prior to delivering NO, a pre-use/high pressure leak test and a low calibration must be done:
   a. Open iNO cylinder valve (notice needle on gauge goes up and indicates tank psi) and then close iNO cylinder valve – verify that needle does not go down, which would indicate a leak
   b. Manually purge and depressurize the NO tank (ensure iNO tank is closed prior to depressurizing)
   c. Turn on DSIR monitor and perform a low calibration and pre-use check – follow prompts on monitor screen
3. Verify O_2 on iNO monitor is similar to FiO_2 set on the blender of the isolette
   a. Note that when even though FiO_2 on the blender of the isolette is set at 100%, ~67 – 77% will be delivered; with the added 2-5 LPM flow to drive NO in-line to the patient the sample line to DSIR monitor will monitor FiO2 to be ~86-92%

Notes

- Due to air entrainment, the TXP-2D will deliver ~67-77% FiO2 when connected to 100% O_2 source
- Gas consumption: (1) E tank cylinder with 2000psi will last ~37min
  - (2000psi)(0.28)/15LPM = ~37min
- A heat moisture exchanger (HME) must be placed between the patient and Phasitron breathing circuit to provide adequate humidification necessary to prevent mucosa from becoming thick and sticky that can cause mucus plugging, leading to decreased chest wiggle and impaired ventilation
- Ensure the patient has an orogastric (OG) or nasogastric (NG) tube placed to prevent abdominal distention from HFV
- Altitude can affect ventilation – settings may require adjustments
- Clean the TXP-2D between patient use with approved cleaners
- The Phasitron Breathing Circuit and HME are for single-patient use only and should be discarded after completing patient transport. DO NOT REUSE.
- A preventative maintenance (PM) check should be done annually for the TXP-2D by a Percussionaire-authorized service center
- Batteries in the digital monitor (DM) of the TXP-2D need to be changed out every 6 months

A.1 Record of Revisions

Revision Number	Revision Date	Comment
0	02-12-18	Initial Release
1	04-20-18	Minor corrections, removed obsolete headers
2	06-08-18	Multiple additions and corrections
3	09-04-18	Multiple additions and corrections
4	01-23-19	Added link to drug profile for all medication citations. Multiple corrections.
5	06-11-19	Medication updates, minor corrections and additions
6	10-10-19	Added various links. Multiple updates and corrections.
7	03-09-20	Added various links. Multiple updates and corrections.
8	03-20-20	Added COVID-19 airway management guidance and other minor updates
9	04-09-20	Added COVID-19 management guidance and other clinical updates
10	06-16-20	Added various links. Multiple updates and corrections.
11	09-18-20	Updated neurological guidance
12	12-16-20	Moved neonatal chapter to seperate manual, Updated guidance
13	04-08-21	Multiple updates and corrections.
14	06-28-21	Added various links. Multiple format updates and corrections.
15	11-04-21	Compete review. Multiple updates and corrections.
15.1	11-29-21	Corrected typographical errors and updated cross references
16	02-14-22	Compete review. Multiple updates and corrections.
16.1	02-23-22	Corrected typographical errors
17	06-27-22	Compete review. Multiple updates and corrections.
18	12-01-22	Compete review. Multiple updates and corrections.
18.1	12-27-22	Removed blank pages
19	05-25-23	Compete review. Multiple updates and corrections.
19	05-23-23	Updated revision date
20	09-05-23	Compete review. Multiple updates and corrections.
21	01-08-24	Compete review. Multiple updates and corrections.
21.1	01-15-24	Minor format changes
22	05-01-24	Compete review. Multiple updates and corrections.
22.1	05-28-24	Minor updates, corrected typos, added missing content
23	01-23-25	

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A.2 Change Log

Page	Date	Comment
11-1	01-23-25	Combined Neo guidance into the Clinical Care Guidance

A.3 Document Conventions

A.3.1 Messages

The following messages should prompt readers to pay special attention to specific points:

NOTE: **Note is used to highlight best practice, a preferred choice, or a key point**

COMPLIANCE: **Compliance is used to highlight compulsory actions for legal or regulatory reasons**

NOTICE **Notice is used to highlight essential information, procedures or concepts. The notice may define or dictate actions required by the reader.**

CAUTION **Caution is used to highlight actions and consequences that may cause a patient complication or adverse event. The caution may state what to do, or what not to do, followed by the likely consequences.**

WARNING **Warning is used to highlight actions and consequences that will likely cause harm to a patient. The warning may state what to do or what not to do, followed by the likely consequences.**

A.3.2 Directives

Directives in this publication use the following definitions:

- Must—used for an obligation
- Must not or prohibited—used for a prohibition
- May—used for a discretionary action
- Should—used for a recommendation

NOTE: **The "weight" that must, prohibited, may and should carry should be kept in mind while performing duties**

A.4 Abbreviations

Abbreviation	Definition
AIIR	Airborne Infection Isolation Room
AIS	Acute Ischemic Stroke
AOC	Administrator on Call
ARDS	Adult Respiratory Distress Syndrome
ASV	Adaptive Support Ventilation
BP	Blood Pressure
BPM	Beats Per Minute
BPS	Behavioral Pain Scale
CPOT	Critical Care Pain Observation Tool
CPR	Cardiopulmonary Resuscitation
CTR	Clinical Transport Review
CVP	central venous pressure
D10W	Dextrose 10% in Water
D25W	Dextrose 25% in Water
D50W	Dextrose 50% in Water
D5W	Dextrose 5% in Water
DASH-1A	Definitive airway sans hypoxia/hypotension on first attempt
DBP	Diastolic Blood Pressure
DCA	Damage Control Airway
DFU	Directions for Use
DSA	Delayed Sequence Airway
DSI	Delated Sequence Intubation
ECG	Electrocardiogram
EFM	Electronic Fetal Monitor
EHR	Electronic Health Record
ETCO2	End-Tidal Carbon Dioxide
ETI	Endotracheal Intubation
ETT	Endotracheal Tube
FHR	Fetal Heart Rate
FiCO2	Fraction of inspired CO2
FiO2	Fraction of Inspired Oxygen
GCS	Glasgow Coma Scale
HMEF	Heat and Moisture Exchanging filter
IBW	Ideal Body Weight
IFT	Interfacility Transfer
IM	Intramuscular
IV	Intravenous
IVP	Intravenous Push
IVPB	Intravenous Piggyback
LR	Lactated Ringer's
MIVF	Maintenance Intravenous Fluids
NA	Non Applicable
NAS	Nasal
NIBP	Non-invasive Blood Pressure
NICHD	National Institute of Child Health and Human Development
NIV	Non-invasive ventilation
NMB	Neuromuscular Blocking Agent
NS	Normal Saline
PEEP	Positive End Expiratory Pressure
POC	Point of Care
PPE	Personal Protective Equipment
PRN	As needed or necessary
QRG	Quick Response Guide
RASS	Richmond Agitation Sedation Scale
REL	Required Equipment List
RFT	Reason for Transport
ROSC	Return of Spontaneous Circulation
RSA	Rapid Sequence Airway
RSI	Rapid Sequence Intubation
SAS	Riker Sedation-Agitation Scale
SBP	Systolic Blood Pressure
SL	sublingual
SMR	Spinal Motion Restriction
SVN	Small Volume Nebulizer
SpCO	Carboxyhemoglobin saturation as measured by pulse CO-oximeter
SpMet	Methemoglobin saturation as measured by CO-oximeter
SpO2	Percent oxygen saturation of arterial blood as measured by pulse oximetry
TBSA	Total Body Surface Area
TBW	Total Body Weight
TOD	Top ofÂ Descent
TPA	Tissue Plasminogen Activator
TXA	Tranexamic Acid
VF	Ventricular Fibrillation
VT	Ventricular Tachycardia

A.5 Clinical Abbreviations

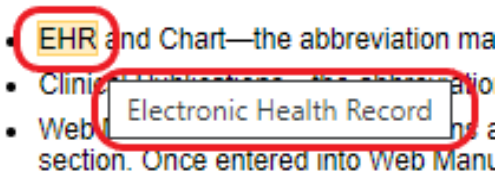
Approved abbreviations may be used for patient care documentation and efficient transfer of information in clinical publications. The abbreviations and acronyms policy and Web Manuals abbreviation module are the definitive source for approved clinical abbreviations.

⚠ CAUTION **Some abbreviations can have multiple meanings or are only used regularly within a given specialty, using them can represent a hazard to the patient if the abbreviation is misinterpreted.**

A.5.1 PHI Acronyms and Abbreviations Policy

Enterprise approved abbreviations are listed in the PHI acronyms and abbreviations policy. Abbreviations in the enterprise section of the policy may be used in all clinical publications such as polices, guidelines and procedures. In addition, any abbreviation in the enterprise section may be used in handwritten or typed patient care documentation.

- Abbreviations in the Enterprises section of the PHI acronyms and abbreviations policy may be hyperlinked in Web Manuals. If the abbreviation is hyperlinked to the definition, it will be highlighted and the definition will appear when the cursor is hovered over the abbreviation.



Supplemental Clinical Abbreviations

The supplemental clinical abbreviations section defines and further specifies additional abbreviations that may be used in all clinical publications such as polices, guidelines and procedures as well as handwritten or typed patient care documentation.

Charting Abbreviations

The charting abbreviations section defines and further specifies additional abbreviations that may only be used in handwritten or typed patient care documentation.

NOTICE **Use of supplemental clinical abbreviations in training materials and clinical documents like guidelines, procedures and drug profiles is prohibited. The word should be spelled out in these documents.**

A.6 Definitions

Definition	Description
Handoff	A handoff is the transfer of patient information, authority and responsibility, from one clinician or team of clinicians to another clinician or team of clinicians.

A.7 Metric Weights and Measures

Metric weights and measures should be used for all clinical documentation, written guidance and verbal communication. The metric directive applies to patient care, medical operations and clinical administrative activities.

- Weight must be measured in kilograms (or grams for newborns)
- Height or length must be measured in centimeters
- Temperature must be measured in degrees Celsius

NOTICE

The use of US standard measures (e.g. feet, inches, pounds, ounces, degrees Fahrenheit, etc.) is prohibited.

NOTICE

The use of apothecary system measures or symbols (e.g., grains, drams, minims, ounces), or household measurements (e.g., teaspoons, tablespoons) is prohibited.

A.7.1 Standard Weights and Measures Abbreviations

The following standard abbreviations should be used for clinical documentation, publications, written records and dosage units.

- °C—degrees Celsius
- cm—centimeter(s)
- gm—gram(s)
- kg—kilogram(s)
- L—liter(s)
- m—meter(s)
- mcg—microgram(s) (do not use µ)
- meq—milliequivalent(s)
- mg—milligram(s) (do not use mgs)
- mL—milliliter(s) (do not use cc, ml, mLs, or mls)
- nanog—nanogram(s)

A.8 Medication Dosing Schemas

Medication doses are determined using the following frameworks

- Fixed
- Ideal body weight (IBW)
- Total body weight (TBW)

NOTE: **Dosing schemas may be different for adult and pediatric patients.**

A.8.1 Standard Fixed Dose

The fixed dose applies a specific dose or dose range that does not vary with patient weight. The drug may be administered as a one time dose or as an infusion titrated to patient response or condition. Standard fixed dose calculations are used primarily in adult patients. Examples of fixed doses include:

- Epinephrine 1 mg IVP
- Hydrocortisone 100 mg IVP
- Norepinephrine 2–35 mcg/min IVPB

A.8.2 Weight Based Dose

Weight based medication administration schemas use the patients weight to determine the dose for administration. Two different weight based scalars are used depending on the medication, TBW or IBW.

Total Body Weight

Total body weight, also known as actual body weight is used to calculate a few specific medications. TBW dosing may be identified by the TBW after the dose/kg:

- Etomidate 0.1 mg/kg **TBW** IVP

Ideal Body Weight

Ideal body weight is the predominant weight based dose calculation method. In adults, the IBW is determined by the patients gender and height using the Devine formula. In pediatric patients, the IBW is based on the Handtevy system which uses the patients age or height.

IBW dosing may be identified by the IBW after the dose/kg:

- Naloxone 0.001–0.01 mg/kg **IBW** IV/IM/IN

NOTE: **If a weight based scalar is not identified, use IBW**
NOTE: **Additional information on determining patient weight may be found in the** Age, Height and Body Weight **guidelines.**

A.8.3 Intramuscular (IM) Route Guide

- Administration volumes and patient age/weight dictate site selection, needle size and suggested maximum volume to be administered in any single site.
- The vastus lateralis, rectus femoris and ventrogluteal sites are preferred for large volume deep muscle IM injections. These injection sites are reliable and accessible when a patient is lying supine.
- The deltoid is suitable for small volume IM injections.

Age/Weight			Vastus Lateralis or Rectus Femoris	Ventrogluteal	Deltoid
	Preemie	2 kg	• Needle: 5/8–7/8 inch • Max volume: 0.5 mL	Not Recommended	Not Recommended
	Newborn	4 kg			
	4 MO	6 kg			
	6 MO	8 kg			
	1 YR	10 kg	• Needle: 5/8–1 inch • Max volume: 1 mL	• Needle: 5/8–1 inch • Max volume: 1 mL	• Needle 5/8–1 inch • Max volume 0.5 mL
	2 YR	12 kg			
	3 YR	15 kg			
	4 YR	17 kg	• Needle: 1 inch • Max volume: 2 mL	• Needle: 1 inch • Max volume: 2 mL	• Needle: 1 inch • Max volume: 1 mL
	5 YR	20 kg			
	6 YR	22 kg			
	7 YR	25 kg			
	8 YR	27 kg			
	9 YR	30 kg			
	10 YR	35 kg	• Needle: 1–1.5 inch • Max volume: 4 mL	• Needle: 1–1.5 inch • Max volume: 4 mL	• Needle: 1–1.5 inch • Max volume: 1.5 mL
	11 YR	40 kg			
	12 YR	50 kg			
	13 YR	60 kg			
	Adult		• Needle: 1–3 inch • Max volume: 5 mL	• Needle: 1–3 inch • Max volume: 5 mL	• Needle: 1–3 inch • Max volume: 2 mL

NOTICE **If more than 1 injection is required due to site volume limits or multiple medications, each injection should be administered in distinctly separate sites e.g., left and right vastus lateralis.**

NOTICE **To prevent inadvertent needlestick injury or reuse, needle safety mechanisms should be deployed immediately after administration.**

A.9 Clinical Publication Management

Patient care guidelines and associated publications such as the Handtevy guide and drug profiles are the primary treatment resources for PHI clinicians. To the greatest extent possible, PHI patient care guidelines are based on up-to-date medical evidence derived from the published literature or PHI clinical performance data.

A.9.1 Publication Review and Revision

Publication Review

To ensure guidance is up to date, all publications are reviewed no less that once per year.

Publication Revision

Clinical publications including guidelines, procedures, action guides and related clinical manuals are updated and revised as change requests are received and approved by the manual owner, DCS or Clinical Practice Committee.

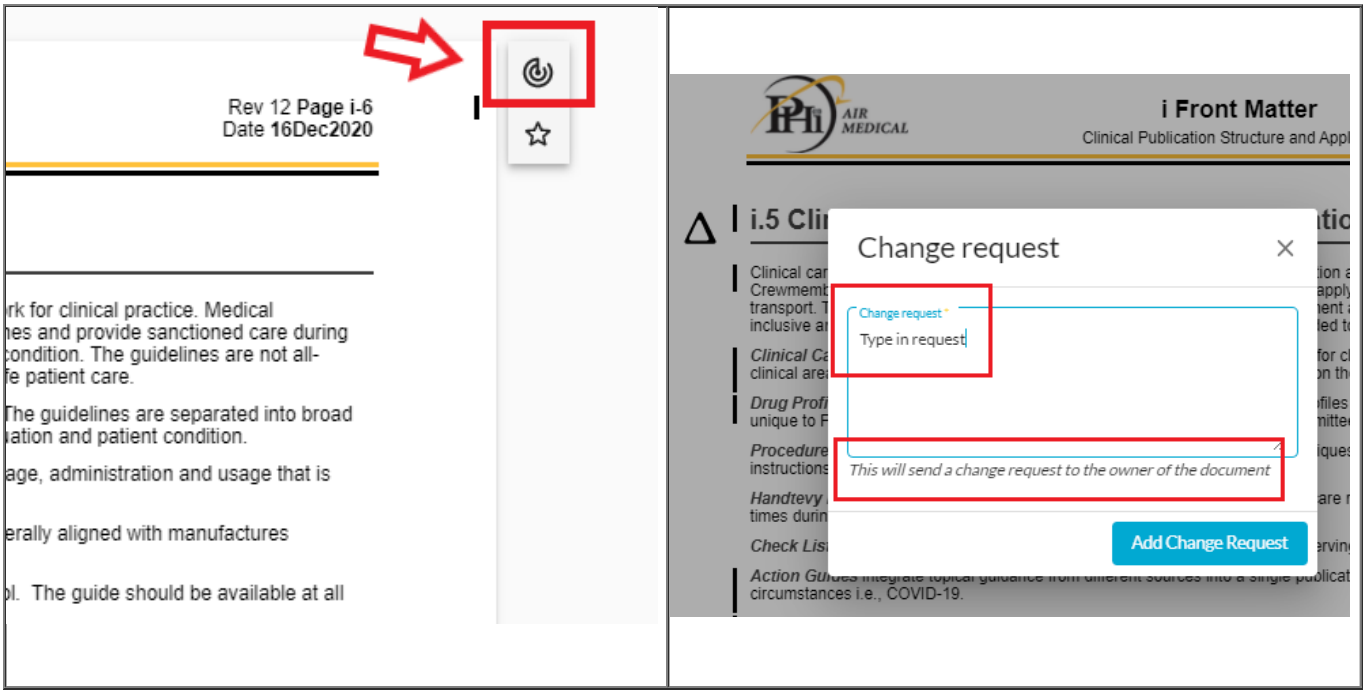
NOTE: **Each time a publication is revised, it is also comprehensively reviewed by the manual owner.**

A.9.2 Web Manuals Change Request

Any manual user may submit a change request.

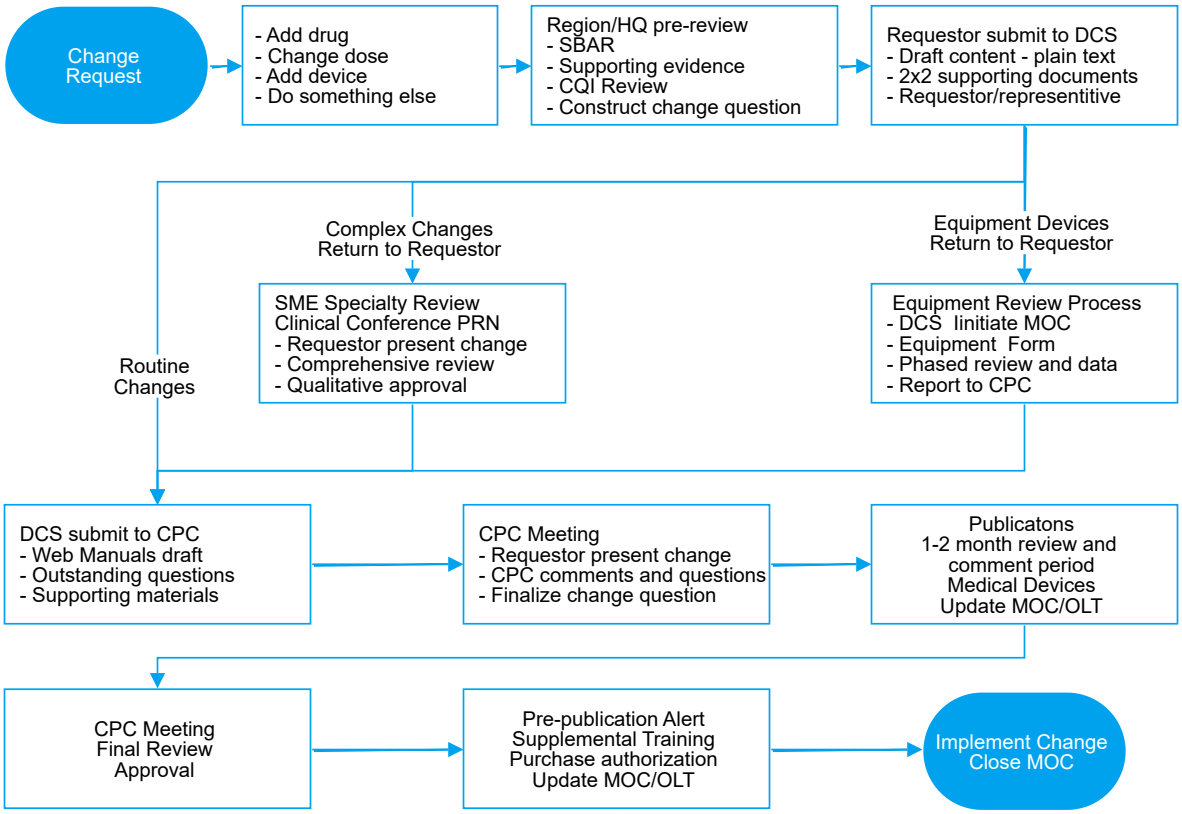
While on the applicable page in the Web Manuals reader:

1. Select the change request icon at the upper right of the page
2. Enter a complete and specific change request in the box
3. Select Add Change Request
4. The change request will be sent to the document owner



A.10 Clinical Change Request

Clinical policy, process, procedure or equipment changes may be initiated by doing one of the following:

- Submit a change request in the applicable Web Manuals document.
- Send an email message to the region clinical manager.

A.10.1 Change Request Flow